VERTIGO

ENT | 3rd Year MBBS

Hi Sammy!
Vertigo = tricky but
very high yield!
Learn the tables!

Vertigo | BPPV | Meniere's | ENT

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Peripheral vs Central Vertigo

This comparison table is THE most important thing in vertigo!

DEFINITION

Vertigo = illusion of movement (rotation). NOT just dizziness!

Peripheral vs Central Vertigo (THE Table)

Feature	Peripheral Vertigo	Central Vertigo
Onset	Sudden	Gradual or sudden
Severity	Severe, intense	Mild to moderate
Duration	Seconds to hours	Days to weeks
Nystagmus	Horizontal/torsional	Vertical/multidirectional
Fixation	Suppressed by fixation	NOT suppressed by fixation
Hearing loss	Common	Rare
Tinnitus	Common	Rare
Neuro signs	Absent	Present (ataxia, diplopia)
Nausea/vomiting	Severe	Variable
Fatigability	Nystagmus fatigues	Does NOT fatigue

Causes of Vertigo

PERIPHERAL (90%)

- BPPV (most common overall)
- Meniere's disease
- Vestibular neuritis
- Labyrinthitis
- Acoustic neuroma

CENTRAL (10%)

- Cerebellar stroke/hemorrhage
- Vestibular migraine
- Multiple sclerosis
- Brain tumors (CPA tumors)
- Vertebrobasilar insufficiency

★ **90-10 Rule!**
90% of vertigo is PERIPHERAL (inner ear) and 10% is CENTRAL (brain). If neuro signs = central!

BPPV = #1 cause of vertigo! Dix-Hallpike is the gold standard!

Benign Paroxysmal Positional Vertigo (BPPV)

Cause:	Otoconia (calcium carbonate crystals) dislodged into semicircular canal
Canal:	Posterior semicircular canal most commonly affected (80-90%)
Triggers:	Head position change — rolling in bed, looking up, bending
Duration:	Brief episodes (< 1 minute), recurrent
Age:	Most common in elderly; may follow head trauma

Dix-Hallpike Test (Diagnostic)

1. Patient sitting, head turned 45 degrees to tested side
2. Rapidly lay patient supine with head hanging 20-30 degrees off table
3. Observe for nystagmus (latency 2-5 sec, crescendo-decrescendo)
4. Positive = upbeat torsional nystagmus + vertigo (< 60 seconds)
5. Nystagmus fatigues on repetition (hallmark of peripheral cause)

Epley Maneuver (Canalith Repositioning — Treatment)

1	Step 1:	Dix-Hallpike position (affected ear down, head hanging)
2	Step 2:	Turn head 90 degrees to opposite side (wait 30 sec)
3	Step 3:	Turn body & head further 90 degrees (nose down, wait 30 sec)
4	Step 4:	Sit patient up slowly with chin tucked down
5	Result:	Repositions otoconia from SCC back to utricle; 80% cure rate

★ **BPPV Nystagmus!**
Latent, Torsional, Fatigable, Brief (< 60s) = PERIPHERAL. No latency, non-fatigable = CENTRAL!

Meniere's TRIAD is
asked in every exam!
Must know cold!

Meniere's Disease (Endolymphatic Hydrops)

Pathology:	Endolymphatic hydrops — distension of membranous labyrinth
Classic Triad:	Episodic vertigo + Tinnitus + Fluctuating SNHL
+ Aural fullness:	Tetrad if you include sensation of ear pressure
Vertigo:	Episodic, lasting 20 minutes to 12 hours, recurrent
Hearing loss:	Low-frequency SNHL initially, progresses with attacks
Treatment:	Low-salt diet + diuretics (first line), betahistine, intratympanic gentamicin

Vestibular Neuritis vs Labyrinthitis

Feature	Vestibular Neuritis	Labyrinthitis
Cause	Viral (HSV-1) inflammation	Viral/bacterial inner ear infection
Vertigo	Severe, acute onset	Severe, acute onset
Hearing loss	ABSENT (only vestibular)	PRESENT (cochlear + vestibular)
Nystagmus	Away from affected side	Away from affected side
Duration	Days to weeks	Days to weeks
Treatment	Vestibular suppressants, rehab	Antibiotics if bacterial + rehab

★ Meniere's Memory!
Meniere's = ELF: Episodic vertigo + Low-frequency hearing Loss + Fullness in ear
Diagnosis of exclusion! Must have 2+ episodes of vertigo lasting 20 min - 12 hrs.

Quick Differentials of Vertigo

Condition	Hearing	Duration	Key Feature
BPPV	Normal	Seconds	Positional, Dix-Hallpike +ve
Meniere's	Low-freq SNHL	Hours	Triad + aural fullness
Vest. neuritis	Normal	Days	Acute, no hearing loss
Labyrinthitis	SNHL	Days	Acute, with hearing loss
Acoustic neuroma	Unilateral SNHL	Variable	Progressive, CN V/VII signs

Caloric Test + Investigations

COWS mnemonic for caloric test = viva gold! Learn it!

Caloric Test (Bithermal Caloric Test)

1. Tests function of lateral (horizontal) semicircular canal individually
2. Cold water (30C) -> nystagmus AWAY from stimulated ear
3. Warm water (44C) -> nystagmus TOWARD stimulated ear
4. COWS = Cold Opposite, Warm Same (fast phase of nystagmus)
5. Canal paresis > 20-25% = abnormal (unilateral vestibular hypofunction)

COWS Mnemonic — Caloric Response

Stimulus	Fast Phase Direction	Interpretation
Cold water (30C)	Opposite ear	Normal response = nystagmus away
Warm water (44C)	Same ear	Normal response = nystagmus toward
No response	None	Dead labyrinth (canal paresis)
Ice water (0C)	Opposite ear	Used in unconscious patients

Other Investigations for Vertigo

Investigation	Purpose
Pure tone audiometry	Hearing assessment — SNHL pattern in Meniere's
Electronystagmography	Records nystagmus — differentiates peripheral vs central
MRI brain + IAM	Rule out acoustic neuroma / CPA tumors
VEMP (Vestibular evoked)	Tests saccule & inferior vestibular nerve
Romberg's test	Fall toward affected side in peripheral lesion
Fukuda stepping test	Patient rotates toward affected side
HINTS exam	Head Impulse, Nystagmus, Test of Skew (emergency)

★ COWS = Cold Opposite, Warm Same!
This tells you the direction of FAST PHASE of nystagmus in caloric testing.
If BOTH cold & warm give no response on one side = dead labyrinth = canal paresis!

Vertigo = 15+ marks
if you nail the tables!
You've GOT this Sammy!

Most Repeated LAQs (10–12 marks)

1. Differentiate peripheral vs central vertigo with causes & features
2. Describe BPPV — pathophysiology, Dix-Hallpike test & Epley maneuver
3. Describe Meniere's disease — triad, diagnosis & management
4. Describe caloric test — procedure, interpretation & clinical significance

Most Repeated SAQs (5–6 marks)

1. BPPV — pathophysiology & Dix-Hallpike test
2. Meniere's disease — classic triad & management
3. COWS mnemonic — explain with caloric test
4. Vestibular neuritis vs labyrinthitis
5. Romberg's test & its significance in vertigo
6. Acoustic neuroma — clinical features & diagnosis

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Most common cause of vertigo	BPPV
Gold standard test for BPPV	Dix-Hallpike test
Treatment of BPPV	Epley maneuver (canalith repositioning)
Meniere's triad	Vertigo + tinnitus + SNHL
Endolymphatic hydrops	Meniere's disease
COWS	Cold Opposite, Warm Same
Vertigo + no hearing loss	Vestibular neuritis
Vertigo + hearing loss (acute)	Labyrinthitis
Caloric test assesses	Lateral semicircular canal function
Positive fistula test	Labyrinthine fistula (cholesteatoma)
Tullio phenomenon	Sound-induced vertigo (SCC dehiscence)
Most common CPA tumor	Acoustic neuroma (vestibular schwannoma)

♥ **The world may spin, but your knowledge stays rock solid, Sammy!**
Balance is everything — in life and in the inner ear!